

Three-Dimensionally Ordered Mesoporous (3DOm) Carbon Materials as Electrodes for Electrochemical Double-Layer Capacitors with Ionic Liquid Electrolytes

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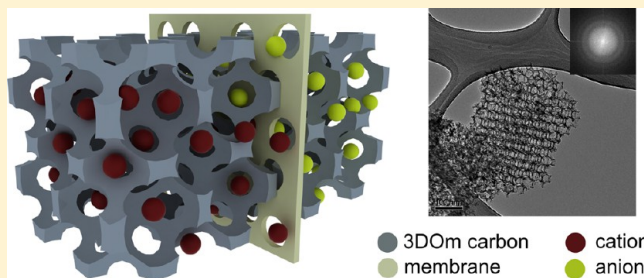
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Supporting Information

ABSTRACT: Compared to rechargeable batteries, electrochemical double-layer capacitors (EDLCs) are normally considered to be higher power but lower electrical energy density charge storage devices. To increase the energy density, one can enlarge the interfacial area between electrodes and electrolyte through the introduction of nanopores and employ electrolytes that are stable over wider voltage ranges, such as ionic liquids. However, due to the relatively high viscosity of ionic liquids and large ion sizes, these measures can result in diminished power performance. Here, we describe the synthesis of carbon electrodes that overcome these limitations and simultaneously provide high specific energies and high specific powers in EDLCs using the ionic liquid EMI-TFSI as an electrolyte. A colloidal crystal templating method was optimized to synthesize three-dimensionally ordered mesoporous (3DOm) carbons with well-defined geometry, three-dimensionally interconnected pore structure and tunable pore size in the range from 8 to 40 nm. To achieve precise control over the pore sizes in the carbon products, parameters were established for direct syntheses or seed growth of monodisperse silica nanospheres with specific sizes, using L-lysine-assisted hydrolysis of silicon alkoxide precursors. Porous carbons were then templated from these materials using phenol–formaldehyde (PF) or resorcinol–formaldehyde (RF) precursors. The pore structures of the nanoporous carbon products were characterized in detail, and the materials were tested as electrodes for EDLCs. Optimal pore sizes were identified that provided a large interface between the electrode and the electrolyte while maintaining good ion transport through the relatively viscous electrolyte. 3DOm PF-carbons with pore diameters in the 21–29 nm range exhibited similar high specific capacitance values ($146\text{--}178\text{ F g}^{-1}$ at 0.5 A g^{-1} , with respect to the mass of carbon in a single electrode) as typical large-scale activated-carbon-based EDLCs but showed significantly better high-rate performance ($80\text{--}123\text{ F g}^{-1}$ at 25 A g^{-1}), a result of the more accessible pore space in which ion diffusion was less restricted.

KEYWORDS: colloidal crystal template, mesoporous carbon, electrical double layer capacitor, high-rate performance, ionic liquid



INTRODUCTION

Electrochemical double-layer capacitors (EDLCs) are increasingly finding applications in consumer electronics, memory back-up systems, industrial power, and energy management due to their high power capability and ultralong cycle life.¹ The devices can function as alternatives or supplements to batteries, and, as energy storage devices, support the advancement of other fields, such as renewable energy, transportation, and portable devices.² For instance, in electric vehicles (EVs) and hybrid electric vehicles (HEVs), EDLCs can be used to provide power during acceleration and recapture energy during braking.

EDLCs can act as intermittent power sources, reducing the stress on the battery during high current processes. This combination is more feasible than using EDLCs or lithium-ion batteries alone in EVs or HEVs because each of them has its own limitations. Batteries are able to store a large amount of energy but suffer from slow power delivery, whereas EDLCs can provide high specific power but can only store a limited

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amount of energy. The differences between batteries and EDLCs are due to the mechanism by which charge is stored and delivered in these systems. Batteries store charge in the bulk material, whereas EDLCs store charge only on electrode surfaces via interfacial capacitance. An EDLC can quickly release its stored energy by discharging of the electrolyte–electrode interface. On the other hand, the charge transfer process in a battery is usually coupled with structural changes of the cathode and the anode materials, which is much slower than the ion exchange process at the electrode surface.

The performance of an EDLC is described by the following equations:

$$C = \frac{\epsilon^* \epsilon_0 A}{d} \quad (1)$$

$$E = \frac{1}{2} CV^2 \quad (2)$$

$$P = \frac{V^2}{4R} \quad (3)$$

where C is the capacitance of the cell, ϵ the dielectric constant of the electrolyte phase in the double layer, ϵ_0 the dielectric permittivity of free space, A the electrode surface area, d the effective thickness of the double layer, E the cell energy, P the cell power, V the cell voltage, and R the cell resistance. According to these equations, for a given cell mass or volume, the energy density and power density can be improved by increasing the electrode surface area, lowering the cell resistance, and using an electrolyte with a wider electrochemical window. To improve the capacitance, traditional planar metal electrodes can be replaced with porous carbon electrodes with higher surface area.

In practice, activated carbons with surface areas up to 2000 m² g^{−1} and fairly high electronic conductivity have been widely used for EDLC applications.^{3–13} The high surface areas are produced by activating carbon using water vapor, carbon dioxide, or KOH at high temperatures. The process usually leads to the formation of carbonaceous materials with wide pore size distributions, ranging from subnanometer micropores to large macropores.¹⁴ The randomness of pore size distribution and the lack of pore ordering limit access of the electrolyte to the micropores and slow down the mass transport of electrolyte inside the active materials. On the other hand, larger macropores waste active volume and therefore reduce the energy density of the material, since the large pores only act as an electrolyte reservoir and have little contribution to the total capacitance. Instead, microporous and mesoporous carbons with narrow pore size distribution and controlled pore structures are desired as electrode materials for EDLCs.

Microporous carbons potentially have the highest capacitance due to their high surface areas. Particularly, some research groups have recently reported on the anomalous high capacitance of materials with subnanometer pores.^{15–19} Counter to the traditional Helmholtz electrical double layer model for EDLCs, which assumes that the energy of an EDLC is stored via adsorption of solvated ions on the electrode surface, the solvation shells of ions are partially removed when the ions are inside very small pores.¹⁸ This results in smaller charge separation between the ions and the pore walls (smaller d in eq 1), leading to an increase in capacitance. However, the accessibility of these subnanometer pores is very limited, which can adversely impact the power capability of the materials. It is

clear that EDLCs with both high energy and high power densities should be built on hierarchically porous materials since micropores are needed for high capacitance, and mesopores (2–50 nm) are necessary for efficient transport of electrolyte into the micropores.^{20–29} This is particularly important for systems using organic or ionic liquid electrolytes, which have wider electrochemical windows but smaller diffusion coefficients than aqueous electrolytes. More importantly, these mesopores must be designed in the best way to facilitate the transport of electrolyte while avoiding significant reductions in the volumetric energy density of the materials.

Using nonaqueous electrolytes such as organic electrolyte solutions and ionic liquids with wider working voltages is an effective way to improve the energy density and power density of EDLCs. These nonaqueous electrolytes can increase the working voltage of an EDLC cell to 3–5 V (organic electrolytes up to 3.5 V, ionic liquids up to 5.0 V, compared to 0.5–1.0 V for aqueous electrolytes), which results in a significant increase of energy and power density, as E and P vary with the square of the voltage window (V).³⁰ To obtain the same enhancement by increasing the surface area of electrode materials, porous materials with extremely small pores (micropores) must be utilized. However, the charge and discharge rates of such materials is limited by the slow mass transport of ions through the micropores. The resulting trade-off of power density for capacitance negates the rate advantage of EDLCs over rechargeable batteries while the energy density of the capacitor would still not approach that of batteries.

This paper describes the fabrication of carbons with well-ordered, interconnected mesopore structures and microporous walls that offer an optimal balance between mass transport and surface area. The capacitive behavior of these carbons is then studied using an ionic liquid as the electrolyte that remains electrochemically stable up to 3.5 V. The carbon materials were prepared via a colloidal crystal templating (CCT) method using assemblies of monodisperse silica spheres as templates, each assembly with a different size ranging from 10 to 40 nm. CCT offers a versatile method to fabricate nanostructured materials with an open, interconnected structure and controlled pore size. The void space between spheres was then infiltrated with a precursor solution for the target electrode material, here a polymeric precursor for carbon. Thermal processing converted the precursor to a solid skeleton that surrounded the templating spheres. Removal of the silica spheres produced an inverted replica of the original colloidal crystal, that is, an “inverse opal” or three-dimensionally ordered mesoporous (3DOM) solid. The inverse replica inherited the face-centered cubic (fcc) symmetry of the templating sphere array. The fcc macropore array was interconnected through windows formed where templating spheres were in direct contact (i.e., at 12 points in an ideal and typical structure), enabling efficient infiltration with an electrolyte. Ordered mesoporous carbon samples with pore sizes ranging from 10 to 40 nm and surface areas as high as 1600 m² g^{−1} were prepared. Electrochemical cells constructed from phenol–formaldehyde-derived 3DOM carbon electrodes showed high specific capacitance, very high rate performance, and good cyclability, making them suitable for high energy density and high power density EDLCs.

■ EXPERIMENTAL SECTION

Materials. Resorcinol (99%), phenol (99%), formaldehyde (aqueous solution, 37 wt %), sodium carbonate, anhydrous ethanol, methanol, tetrahydrofuran, polytetrafluoroethylene (PTFE, 60 wt %

dispersion in water), tetraethylorthosilicate (TEOS, 99%), L-lysine (98 wt %), hydrochloric acid (0.1 M), potassium hydroxide (98 wt %), sodium hydroxide, and hydrochloric acid (37 wt %) were purchased from Sigma-Aldrich and used without further purification. The ionic liquid 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMI-TFSI, >99%, water content <100 ppm) was purchased from IOLITEC. High surface area activated carbon powder (AB-S20, bulk density 0.32–0.40 g cm⁻³, specified by the supplier for preparing electrodes for high power supercapacitors), was purchased from MTI Corporation. Deionized water was purified to a resistivity greater than 18 M Ω -cm.

Preparation of Silica Spheres. Silica spheres (10–40 nm) were prepared by hydrolysis of TEOS in the presence of L-lysine at different temperatures. Spheres with diameters >20 nm were synthesized by seed growth from the smaller spheres. In general, L-lysine (70 mg) was dissolved in 70 mL water in a 125 mL polytetrafluoroethylene (PTFE) bottle and stirred for 10 min. TEOS (5.4 g) was added into the L-lysine solution and vigorously stirred at room temperature for 30 min before being transferred into an oil bath and stirred at the designated temperatures for different times, as described in the following. SiO₂ spheres with average diameters ~10 nm were synthesized by stirring the mixed solution at 500 rpm for 48 h at 70 °C. SiO₂ spheres with diameters ~14 nm were synthesized by stirring the mixed solution at 500 rpm for 48 h at 90 °C. SiO₂ spheres with average diameters ~21 nm were prepared using seeded growth from 14 nm spheres by adding a total of 10.8 g of TEOS over five separate fractions, each 1 h apart, and stirring at 500 rpm for an additional 48 h at 90 °C. Spheres with 26 or 29 nm diameters were produced following the same steps but starting from seeds with 21 or 26 nm diameters, respectively. SiO₂ spheres with diameters ~40 nm were prepared using seeded growth from ~21 nm spheres, starting with one-third of the volume of the original 21-nm sphere solution and adding three batches of 10.8 g of TEOS each and stirring at 500 rpm at 90 °C for 48 h for each batch. Each of the three batches was divided into 5 fractions (i.e., 2.16 g TEOS per fraction) that were added in 1-h intervals. At the end of each synthesis, the dispersions were then poured into Petri dishes and placed into an oven at 70 °C to allow the spheres to settle and to remove water. The obtained solids were calcined at 550 °C for 6 h in air to remove any organic components and produce silica colloidal crystal templates. All of the above diameters refer to as-synthesized spheres; sphere sizes were 5–10% smaller after calcination.

Preparation of Carbon Precursors. Two different carbon precursors, namely, poly(phenol–formaldehyde) (PF) and poly(resorcinol–formaldehyde) (RF) were used to prepare PF carbon and RF carbon, respectively. The RF precursor was freshly prepared for each batch of carbon, whereas the PF precursor was prepolymerized at 70 °C for 1 h, then dispersed in THF and kept in a refrigerator as a stock solution. For the RF precursor, resorcinol (3.3 g) and sodium carbonate (0.06 g) were dissolved in 4.5 mL aqueous solution of formaldehyde (37 wt %) and vigorously stirred at room temperature for 20 min before being infiltrated into the silica colloidal crystal template. For the PF precursor, phenol (61 g) was melted at 50 °C in a 500 mL round-bottom flask and reacted with 13.6 g NaOH aqueous solution (20 wt %). Aqueous formaldehyde solution (200 mL, 37 wt %) was added dropwise into the phenol solution while stirring. After that, the mixture was heated at 70 °C for 1 h to increase the extent of polymerization between phenol and formaldehyde. The product was neutralized with 30 mL HCl (0.6 M), and water was removed from the mixture by rotary evaporation at 50 °C. The viscous, water-free polymer was dispersed in THF to obtain a 50 wt % dispersion. The polymer dispersion was then filtered to remove NaCl and stored in a refrigerator before use.

Preparation of 3DOm Carbon. 3DOm carbon was prepared by infiltrating carbon precursors into silica colloidal crystal templates for 4 h by capillary action, then aging the infiltrated composites at designated temperatures (85 °C for RF and 140 °C for PF) for 24 h. The aged composites were then pyrolyzed at 900 °C for 4 h (5 °C/min ramping rate) to transform the polymerized PF and RF precursors into carbon. Silica spheres were then etched out by hydrothermal treatment with concentrated KOH (6 M) at 180 °C over 48 h. The

obtained carbon was neutralized with HCl solution (1 M) and washed with copious amounts of water to remove any remaining inorganic salts. The 3DOm carbon samples were denoted PF_n or RF_n, where *n* refers to the diameter of the as-synthesized templating silica spheres in nanometers.

Product Characterization. Raman spectroscopy was performed with a Witec Alpha300R confocal Raman microscope using 514.5 nm incident radiation at the lowest possible beam power to minimize beam damage of the sample. Scanning electron microscopy (SEM) was carried out with a JEOL-6700 microscope operating at 1.5–5 kV with emission currents ranging from 2 to 10 μ A. Because the carbon samples were sufficiently conductive, none of these samples were metal coated. Transmission electron microscopy (TEM) images were obtained with a Technai T12 microscope operating at 120 kV with emission currents ranging from 7 to 12 μ A. Nitrogen sorption isotherms were obtained using a Quantachrome Autosorb iQ. Samples were outgassed for 12 h at 150 °C at 1 milli Torr. Brunauer–Emmett–Teller (BET) surface areas were calculated from the relative pressure range 0.01 < *P*/*P*₀ < 0.2 of the adsorption branch for the commercial activated carbon and the range 0.01 < *P*/*P*₀ < 0.3 for the 3DOm carbon materials. Total pore volumes were calculated from the adsorption branch using the Barrett–Joyner–Halenda (BJH) method. A Quenched Solid State Functional Theory (QSDF) model optimized by Quantachrome for 3DOm carbon was also used to calculate surface areas and surface area contributions associated with specific pore sizes.

Cell Preparation and Electrochemical Tests. For the preparation of electrodes, the 3DOm or activated carbon materials were mixed with 5 wt % polytetrafluoroethylene (PTFE) and extensively ground in a mortar to form a homogeneous mixture. The mixture was rolled into thin films with a pasta-maker, then cut into 0.8-cm diameter circles and used as electrodes. Masses of typical electrodes were 1.7–2.5 mg and thicknesses 120–171 μ m. A typical mass loading of an electrode was about 3.4–5.0 mg/cm². In addition, electrodes of different masses and thicknesses ranging from 0.5 to 6.2 mg were prepared to study the dependence of cell performance on the total mass of electrode material inside the cell. To prepare electrochemical cells, two carbon electrodes of similar mass cut from the same film were first prewetted with electrolyte, separated with a Celgard-3501 membrane, and then packed into a two-electrode cell. The ionic liquid EMI-TFSI was used as the electrolyte. All cells in this study were homemade and based on symmetrical two-electrode EDLC designs (Supporting Information (SI) Figure S1), similar to what has been described as the best-practice method to determine the electrode performance for EDLCs.³¹ Each cell contained two aluminum caps, one of them with a spring-loaded tube to apply pressure on the electrode in the assembled cell. The current collectors were 1 mm thick aluminum disks whose surfaces were polished with superfine alumina powder (0.05 μ m) to minimize the cell resistance and the capacitance contribution from the current collectors.

Electrochemical tests were performed using an Arbin ABTS-2043 test system. Galvanostatic conditions were used to measure the capacitance and to test the cyclability of the EDLCs. For capacitance measurements, the cells were charged and discharged at different rates for 10 to 20 cycles. For cyclability tests, the cells were charged and discharged at a current of 10 mA for a few thousand cycles. The specific capacitance of the electrode material was calculated using eq 4, assuming that the masses and specific capacitances were identical for the two electrodes in the symmetrical cell. Although it is possible that the specific capacitance of the carbon can be different when positively or negatively polarized, the electrodes were cut from the same film with the same mass, and the sizes of cation and anion in the EMI-TFSI electrolyte were similar,¹⁹ making the assumption reasonable.

$$C_{\text{material}} = \frac{4I \cdot \delta t}{m \cdot \delta V} \quad (4)$$

Here, *I* is the discharge current, δt is the time for the cell to discharge, starting at the voltage determined by the maximum voltage (3.5 V) minus the voltage due to ohmic drop, δV is the maximum voltage minus the voltage due to ohmic drop, *m* is the total mass of carbon in

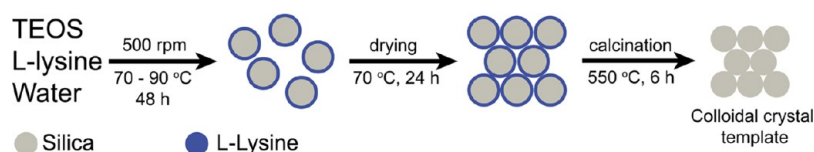


Figure 1. Synthesis scheme of the preparation of silica spheres and colloidal crystal templates.

the two electrodes inside the cell. The current density was calculated by dividing the absolute current by the mass of a single electrode. The specific energy of the symmetrical two-electrode cell was calculated using equation:

$$E_{\text{cell}} = \frac{C_{\text{material}}}{8} V_{\text{max}}^2 \quad (5)$$

where V_{max} is the maximum cell voltage.

Electrochemical impedance spectroscopy (EIS) tests were carried out using a Solartron 1260 frequency analyzer in the frequency range from 10 mHz to 100 kHz at the resting potential with a 20 mV AC amplitude. The gravimetric capacitance of the electrode, C (F g^{-1}), was calculated according to

$$C_{\text{material}} = \frac{2}{2\pi f \text{Im}(z)m} \quad (6)$$

where f is the operating frequency (Hz), $\text{Im}(Z)$ is the imaginary component of the total device impedance (in Ω), and m is the mass of carbon in each electrode (in units of g).

Cyclic voltammetry (CV) curves were obtained with a Solartron 1287 electrochemical station using homemade software. CV curves were scanned at voltage ramp rates from 20 to 100 mV/s. Cell capacitance values were calculated from the CV curves by dividing the current by the voltage scan rate.

RESULTS AND DISCUSSION

One of the main goals of this study was to determine the dependence of capacitance on the pore size of mesoporous carbon electrodes suitable for use with ionic liquid electrolytes. As noted in the introduction, ionic liquids are desirable for EDLCs because of their high working voltages, for example, up to 4.0 V for the EMI-TFSI electrolyte used here, resulting, in theory, in higher energy densities compared to aqueous electrolytes. However, the viscosity of ionic liquids is higher than that of aqueous electrolytes and the ionic conductivity is lower, given the small diffusion coefficients of the bigger ions in an ionic liquid electrolyte. As a result, the rate capability of the EDLC cell may be limited. In order to maintain high rates together with high energy densities, a porous carbon electrode with highly interconnected pores is required. Furthermore, an optimal pore size may exist, which maximizes the available surface area for the double layer, while maintaining optimal ion transport. 3DOM carbon, prepared by hard templating with silica colloidal crystals, has a suitable structure to provide these features, because the mesopores in 3DOM carbon are much larger than the ion sizes, yet small enough to provide large interfaces, and the ordered, interconnected pore structure facilitates mass transport of the electrolyte throughout the electrode. This study shows that for an ionic liquid electrolyte and 3DOM carbon materials with pores sizes in an optimal range, high specific capacitance values can be maintained at high charge and discharge rates.

Preparation of Silica Spheres and Silica Templates. In order to ensure that the pore system in the carbon electrode is well interconnected, it is critical to use highly monodisperse silica spheres with controllable sizes as templates. Mono-disperse colloidal silica spheres readily pack into ordered

structures in which each sphere forms multiple contact points with neighboring spheres. These contact points produce the windows between neighboring mesopores in the templated carbon products. The pore size is determined by the size of each sphere while the ordered pore structure is inherited from the order of the templates. To achieve pores in a size range that can readily accommodate the electrolyte components and also provide high surface areas, silica spheres in the nanometer to tens of nanometer size range are suitable as templates.

The preparation of periodic arrays of suitable subhundred nanometer silica spheres assisted by amino acids was first reported by Yokoi et al. and is outlined in Figure 1.³² The sizes of silica spheres are affected by many factors, such as the concentration of the silica precursor (TEOS), temperature, and stirring rate.³³ In general, higher initial TEOS concentrations, slower stirring, and higher temperatures result in bigger spheres. For the target sphere size for this study (10–40 nm), the sizes of the smallest spheres could be adjusted through the hydrolysis temperature, so that 10 nm spheres were produced at 70 °C and 14 nm spheres at 90 °C.

Larger spheres had to be prepared by a seed growth method, in which additional fractions of TEOS were added to dispersions of preformed silica spheres and heated for various times. The sizes of the product spheres can be related to the sizes of the seed spheres and the relative amount of TEOS added according to the equation:

$$\frac{m_{\text{TEOS(S)}}}{m_{\text{TEOS(F)}}} = \frac{N_{\text{S}} \left(\frac{R_{\text{S}}}{R_{\text{F}}} \right)^3}{N_{\text{F}}} \quad (7)$$

where m_{TEOS} is the mass of TEOS, N is number of spheres in the solution, R is the average radius of each sphere, and d_{sphere} is the density of each sphere. The subscripts S and F refer to seed and final material. This relationship assumes that the total mass of spheres in the solution is proportional to the mass of TEOS used and that the densities of the seed spheres and the final spheres are the same. This is justifiable by the fact that the spheres are all formed from the hydrolysis of TEOS at the same temperature and in the presence of excess water. If the number of spheres does not change ($N_{\text{S}} = N_{\text{F}}$, which means that added TEOS only forms layers outside of the seeded spheres, not any new spheres), it may be expected that the sphere size varies as the cube root of the mass fraction of the two TEOS precursors that were used to produce the two sphere sizes. As shown in the Supporting Information (Table S1 and Figure S2), average sphere diameters estimated from TEM images coincided with theoretical sizes based on eq 7, confirming the predictive value of this equation. Guided by this relationship, highly monodisperse silica spheres of different sizes ranging from 10 to 40 nm were prepared by direct syntheses (Figure 2 A and B) and seeded growth (Figure 2C, D, E, and F).

After evaporation of water, the spheres were packed into ordered face-centered cubic arrays, which can be seen in Figure 3. The small spheres (10 nm) are usually less monodisperse and therefore harder to pack into ordered structures. However,

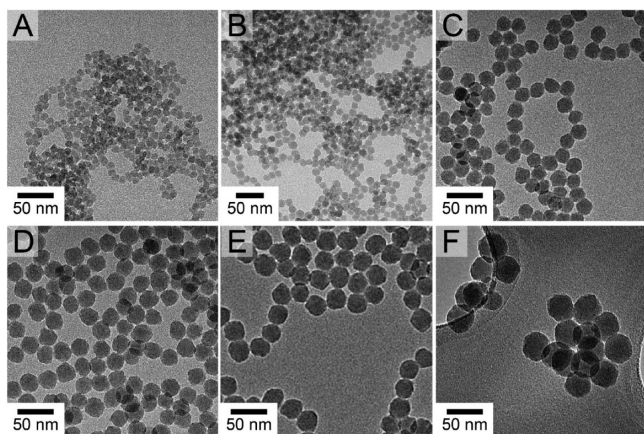


Figure 2. TEM images of (A) 10 ± 1 nm, (B) 14 ± 1 nm, (C) 21 ± 2 nm, (D) 26 ± 2 nm, (E) 29 ± 2 nm, and (F) 40 ± 2 nm SiO_2 spheres. Samples were taken from the solution before calcination.

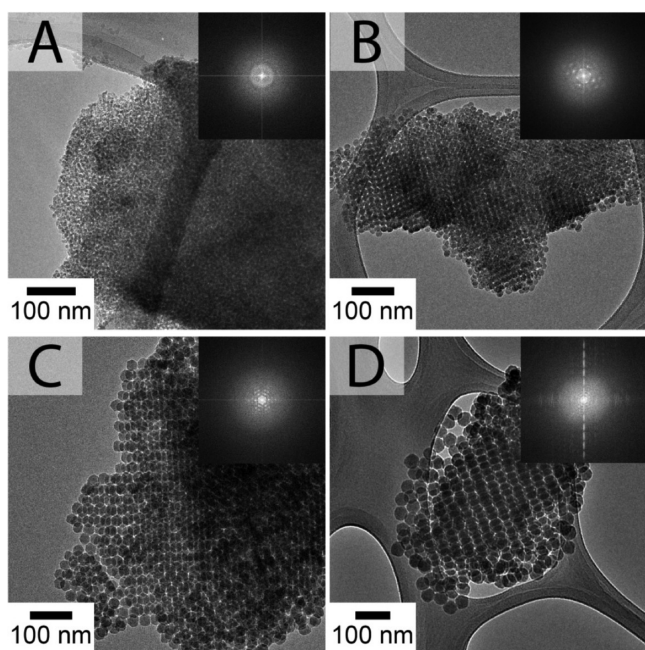


Figure 3. TEM images of (A) 10 ± 1 nm, (B) 26 ± 2 nm, (C) 29 ± 2 nm, and (D) 40 ± 2 nm calcined SiO_2 spheres showing ordered structures. Insets show the corresponding FFT patterns.

the fast Fourier transform (FFT) of the TEM image of the 10 nm silica spheres still shows a ring pattern with a d -spacing of about 10 nm. The bigger spheres (26 and 40 nm) with higher monodispersity are packed into very ordered structures, confirmed by the bright spots in the FFT patterns (Figure 3B and C).

The close-packed silica sphere arrays were calcined in air to remove organic components before being used as templates to prepare 3DOM carbon. All spheres experienced some shrinkage after calcination, which can be ascribed to the loss of organic components and further condensation of silica. The shrinkage varied from 5 to 10% for spheres of different sizes.

Preparation of 3DOM Carbon. The synthesis of 3DOM carbon is outlined in Figure 4. Two different carbon precursors, namely RF and PF, were used. These precursors differ in the number of hydroxyl groups attached to the aromatic ring and, therefore, produce carbons with different oxygen content and surface concentration of functional groups. It is known that carbons with a higher content of oxygen usually have lower electronic conductivity. To study the effects of pore size on the performance of the EDLC electrodes, four different sphere sizes (10 ± 1 , 21 ± 2 , 29 ± 2 , and 40 ± 2 nm) were selected for the templates to prepare 3DOM carbons. First, carbon precursor was infiltrated into the silica template by capillary forces. Both precursors were readily infiltrated into the silica templates. During infiltration, bubbles evolved from the templates, which indicated that these were well wetted by the carbon precursors. The bubbles surrounding the spheres could block access of the precursors into the template and slow down the infiltration; therefore, they needed to be removed, for example by disturbing the solution with a stirring rod. Because the infiltration usually took a long time, it was important to control the gelation of the carbon precursors. RF sol, with its larger number of hydroxyl groups, gels faster than PF sol, which limits its usable time to 6–8 h after the precursor is freshly made. To prolong the gelation time, it was necessary to conduct the infiltration in capped vials, because oxygen from the air catalyzes the polymerization of phenolic compounds and formaldehyde. The composites were then aged at a moderate temperature (85°C for RF, 140°C for PF) to further polymerize the carbon precursor and strengthen the polymer network surrounding the silica spheres. This step also had to be carried out in a closed vial and was necessary to allow most formaldehyde to react with phenol groups, as it is quite volatile and would have been lost if the composite had been pyrolyzed directly after infiltration. After aging at the designated temperatures, both precursors produced robust, dark-red polymer-silica composites, which were transformed into carbon-silica composites by pyrolysis at 900°C under an inert atmosphere.

One advantage of using silica hard templates is that the template provides continual support for the carbon framework during carbonization, preventing structural collapse. SEM images of the carbon-silica composites show that the ordered structure was maintained when the carbon precursor occupied the interstitial space between spheres (SI Figure S3). TEM images of the carbon-silica composites show similar features with the periodic structure of the silica template surrounded by

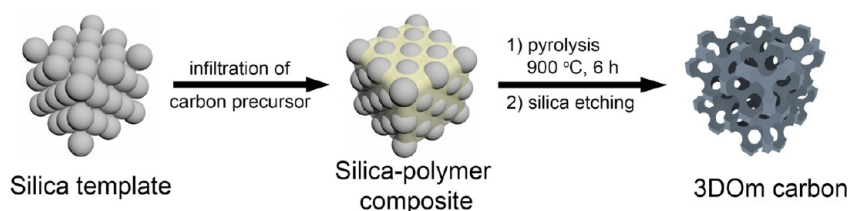


Figure 4. Synthesis scheme for 3DOM carbon.

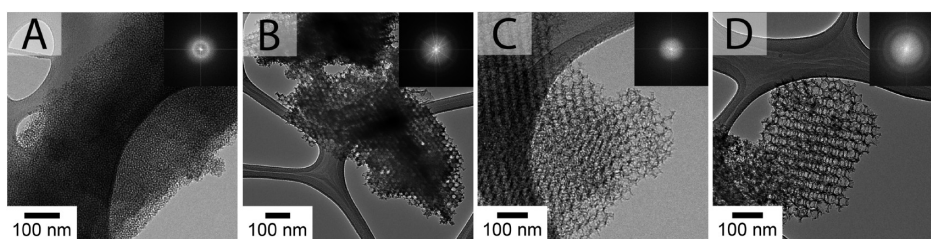


Figure 5. TEM images and corresponding FFTs of 3D carbon (A) PF_10, (B) PF_21, (C) PF_29, and (D) PF_40 carbons.

the carbon phase (SI Figure S4). The carbon phase is very thin, and the spheres are in close contact with each other, which is critical for the formation of windows that connect pores once the spheres are removed. At this stage, the spheres have already shrunk compared to the spheres in the solution due to the condensation of silica at high temperature. Shrinkage was estimated from these TEM images to be about 2–3 nm for both 29 and 40 nm spheres. The silica template was then etched out using a hot, concentrated KOH solution, creating highly ordered mesoporous carbons. In addition to carbon, the final products also contained hydrogen and oxygen (ca. 0.7 wt % H and 9–11 wt % O for PF-derived 3D carbon, ca. 2.0 wt % H and 29 wt % O for RF-derived 3D carbon).

3D Carbon Morphology. The morphologies of 3D carbon prepared from the PF and RF precursors are quite similar when viewed under the TEM. However, the sizes of ordered domains in PF carbons are normally bigger than those in RF carbons. All carbon samples except for the sample templated from the smallest spheres (10 nm) showed ordered structures inherited from the templates (Figure 5 and SI Figure S5) that could be indexed to an fcc geometry on the basis of FFTs of TEM images taken at multiple angles (SI Figure S6). Other structures, such as hexagonally close packed or disordered structures, could also be found in different locations of the sample. However, they only occupied a small fraction of the sample (ca. 5%). Although PF_10 was less ordered, an FFT image of this sample showed a ring pattern with a d -spacing close to the diameter of the templating spheres (Figure 5A), which indicates that the carbon was indeed templated and had an ordered structure but over smaller domain sizes. The larger-sphere templates produced carbon with larger windows and thicker walls, which can in turn accommodate micropores that raise the surface area of the material and could be advantageous for EDLC materials. It is known that micropores in PF or RF carbons are formed from the removal of water and small organic molecules during the aging and the pyrolyzing steps.

Raman spectroscopy was used to characterize the extent of graphitization of the 3D carbon samples. The Raman spectra of all 3D carbon samples showed two prominent peaks associated with sp^2 carbon (G-band, 1597 cm^{-1}) and sp^3 carbon (D-band, 1351 cm^{-1} ; see SI Figure S7). The G-band originates from the in-plane bond stretching motion of pairs of sp^2 carbon atoms, whereas the D-band involves a breathing mode of A_{1g} symmetry, which is only active in the presence of disorder.³⁴ The G peak shifts from 1581 cm^{-1} in graphite to 1595–1599 cm^{-1} in 3D carbon, which indicates that carbon prepared from RF and PF precursors contains mostly nanocrystalline graphite. The domain sizes can be estimated using the equation $I_D/I_G = C(\lambda)/L_a$, where I_D , I_G are the intensities of the disordered peak and the graphitic peak, respectively, $C(\lambda = 515.5 \text{ nm}) = 4.4 \text{ nm}$, λ is the wavelength of the laser used in the Raman spectrometer, and L_a is the cluster

diameter of nanocrystalline graphite.³⁵ To improve the accuracy of the I_D/I_G value, the spectra were deconvoluted with four Gaussian peaks situated at 1190, 1351, 1518, and 1597 cm^{-1} , where the additional bands at 1190 and 1518 cm^{-1} are associated with sp^3 carbon vibrations. The graphitic domain sizes of both PF and RF carbon are quite small (3.6–4.0 nm), which is commonly seen for carbons templated from resol precursors. It is noteworthy that the graphitic domain sizes increase slightly with increasing size of the templating spheres (SI Figure S7, right).

Pore Size Analyses. To understand the interesting electrochemical behavior of 3D carbon electrodes discussed below, it will be necessary to have a description of the pore architecture not only at the level of mesopores introduced by the template but also of micropores that can be present in these types of carbons as a result of the synthesis. Nitrogen adsorption analysis was used to study the surface areas, pore size distributions, and pore volumes of the 3D carbon samples. All 3D carbon samples showed type IV isotherms and type H1 hystereses, with the loops created from adsorption and desorption branches being almost vertical and parallel (Figure 6). The pore size distributions estimated from the adsorption branches using a BJH model show pores that are slightly smaller than the primary silica template particles, as determined from TEM images of as-synthesized spheres. Pore sizes of 7.2, 18.5, 23.6, and 32.6 nm were estimated for 3D carbon PF carbons templated from 10, 21, 29, and 40 nm silica spheres, respectively. The difference is due to shrinkage of the silica spheres during calcination and possibly shrinkage of the mesopores after removal of the supporting silica spheres. In addition, the errors resulting from the application of the BJH model, which was originally developed for cylindrical pores,³⁶ to 3D carbon with spherical pores may also contribute to this reduction in the reported pore sizes. The surface areas of all PF carbons (Table 1) are significantly higher than the surface areas calculated for 3D carbon structures with only mesopores, whose sizes were estimated from the BJH model, strongly suggesting the presence of micropores in these samples. However, the t -method failed to estimate the micropore surface area of these samples, which once again is a consequence of the differences between the cylindrical pore model and spherical pores present in 3D carbon. The nitrogen sorption isotherms of 3D carbon RF carbons are very similar to those of 3D carbon PF carbons. For samples templated from silica spheres of similar size, the predicted surface areas, pore size distributions, and pore volumes using the same models are also very similar (Table 1). The isotherm of the commercial activated carbon used here for comparison is typical for a microporous material, with a sharp increase of adsorbed volume at a very low pressure and a plateau over a wide range of pressure. The surface area of the activated carbon is very high (2023 $m^2 g^{-1}$), most of which is attributed to micropores.

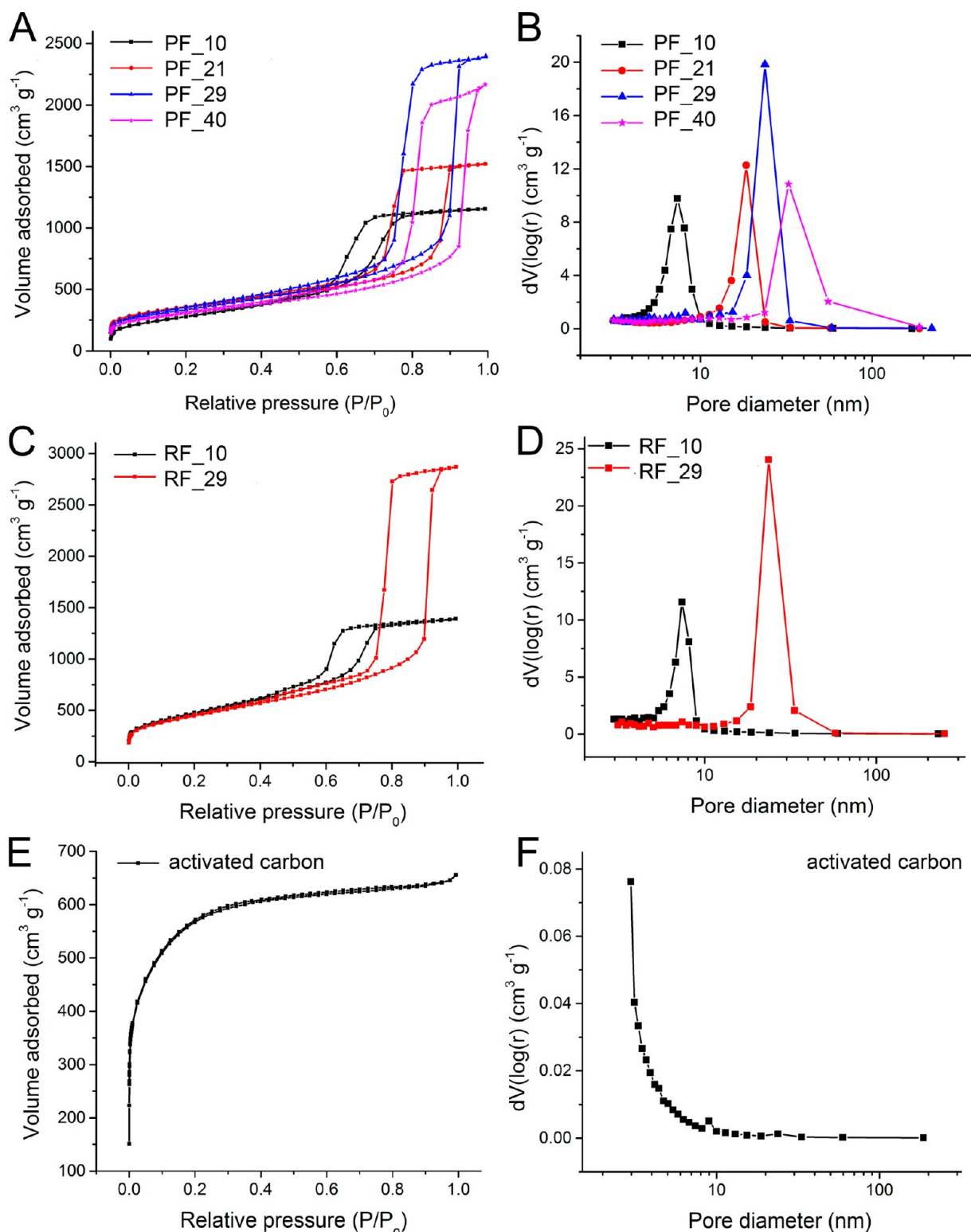


Figure 6. (A) Nitrogen adsorption isotherms and (B) pore size distributions of 3D Om PF carbons templated from silica spheres of different sizes. (C) Isotherms and (D) pore size distributions of 3D Om RF carbons templated from silica spheres of different sizes. (E) Isotherms and (F) pore size distribution of commercial activated carbon.

Recently, a Quenched Solid State Functional Theory (QSDF) model was developed to simulate the adsorption/desorption of nitrogen in spherical pores.³⁶ QSDF-derived histograms of surface area vs pore size of the four 3D Om PF carbon samples are shown in Figure 7 and reveal additional fine structure in the pore size distribution. For PF₁₀ carbon with

thin walls, a single peak around 10.6 nm indicates that the sphere-templated mesopores contributed mostly to the surface area of this sample. The other samples show contributions to surface area from both micropores and mesopores, the largest fraction of micropores being observed in 3D Om PF₄₀, which had the largest mesopores and the thickest walls.

Table 1. Nitrogen Adsorption Data for 3DOm PF, 3DOm RF, and Activated Carbons

sample name	BET surface area (m ² /g)	QSDFT surface area (m ² /g)	BJH pore size (nm)	BJH pore vol. (cm ³ /g)
PF_10	1019	923	7.2	1.75
PF_21	1249	1172	18.5	2.01
PF_29	1171	1196	23.8	3.50
PF_40	1065	1004	32.6	3.08
RF_10	1653	1506	7.4	1.8
RF_29	1588	1414	23.6	4.08
activated carbon AB-520	2023	1447	2.9	0.12

Electrochemical Performance—Galvanostatic Tests.

We compared the electrochemical performance of 3DOm PF carbons templated from spheres with four different sizes, 3DOm RF carbons from two differently sized spheres, and activated carbons, using EMI-TFSI as the ionic liquid electrolyte. Although EMI-TFSI can be used at voltages higher than 4.0 V, in this study, the cell voltage was limited to 3.5 V to avoid electrolyte decomposition due to oxidation.³⁷ The galvanostatic charge and discharge data for 3DOm PF carbons showed triangular shapes typically associated with capacitive behavior (Figure 8A). In most cases, the first charge cycle showed curvature at a cell voltage higher than 2.7 V, which we attribute to the irreversible oxidation of oxygen-containing functional groups³⁸ on the surface of the carbon materials. The feature gradually disappeared in the following cycles. All 3DOm PF carbons showed small ohmic drops (~20–50 mV), confirming the good conductivity of the carbon samples

(Figure 8B). The specific capacitance of 3DOm PF carbons ranged from 146 to 178 F g⁻¹ at a current density of 0.5 A g⁻¹. At a current density of 5.0 A g⁻¹ these values were in a similar range and decreased to ~90% after 1000 cycles (Figure 9). These values are much higher than the capacitances of commercial activated carbons, which are typically less than 100 F g⁻¹.^{39–41} The specific capacitance of 3DOm PF carbons is greater than that of microporous carbide-derived carbons^{19,42} and comparable to that of hierarchical porous carbide derived carbons and activated graphene.^{20,21,43} The specific energy of cells made from two identical 3DOm PF carbon electrodes approached 64 Wh/kg (based on a capacitance of 150 F g⁻¹ and a working voltage of 3.5 V). The practical value is estimated to be 19 Wh/kg based on a weight ratio of 30% for the active electrode material typical for large-scale activated carbon-based EDLCs.

The galvanostatic cycling data of 3DOm RF carbons showed similar features as those of 3DOm PF carbons with triangular shapes, with a slight curvature at higher voltage values during the charging step for the first few cycles (SI Figure S8). This curvature is more prominent in RF carbons, particularly in 3DOm RF carbon with small mesopores (7.4 nm). The ohmic drops for 3DOm RF carbons (30 mV for RF_10 and 62 mV for RF_29) are significantly higher than those of 3DOm PF carbons. These features are expected since the RF carbons have a higher oxygen content and lower conductivity than the PC carbons. The capacitance values of 3DOm RF carbons are quite low (<30 F g⁻¹) compared to those of PF carbons.

For comparison, galvanostatic tests were also carried out with a commercially available, high-surface-area, activated carbon. The charge and discharge curves also formed the triangular

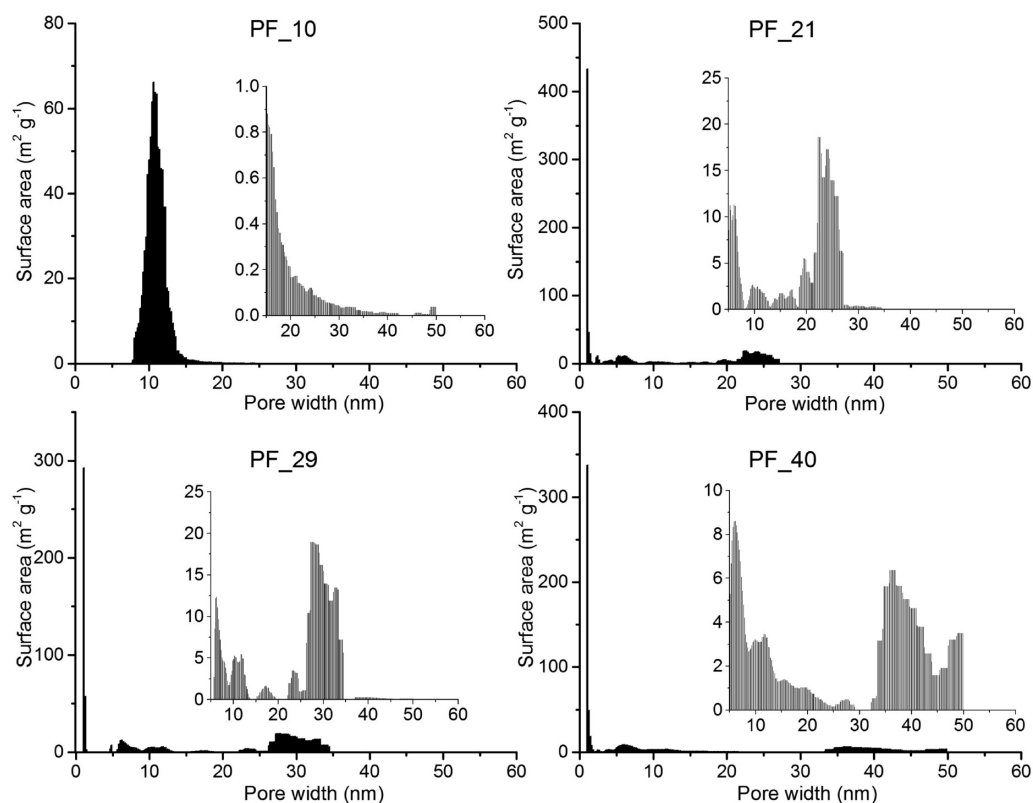


Figure 7. Histograms of surface area vs pore size of 3DOm PF carbon templated from silica spheres with different sizes. The insets show expanded views of the histograms, which were derived by QSDFT methods.

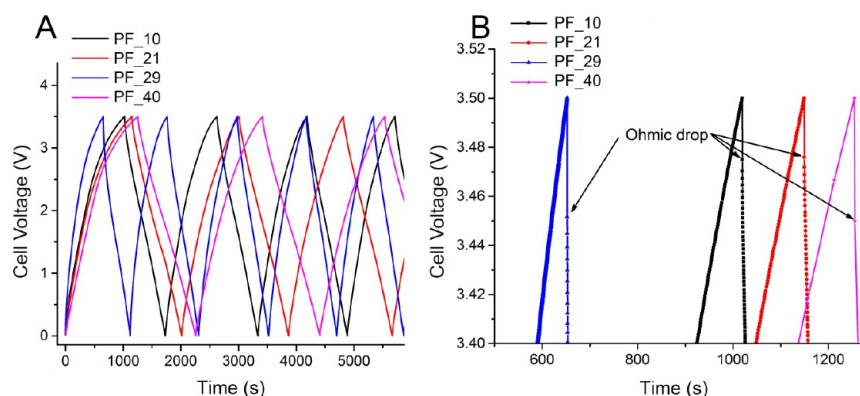


Figure 8. (A) Galvanostatic charge and discharge data of 3D Om PF carbons showing triangle shapes and indicating good capacitive behavior. (B) Zoomed-in area at the highest cell voltages, showing low ohmic drops, that is, the first recorded voltage after switching from charging to discharging. All cells were cycled at a current of 1 mA. This corresponds to 0.5 A g^{-1} current density, based on the mass of active material in a single electrode, both electrodes in the cell having identical mass.

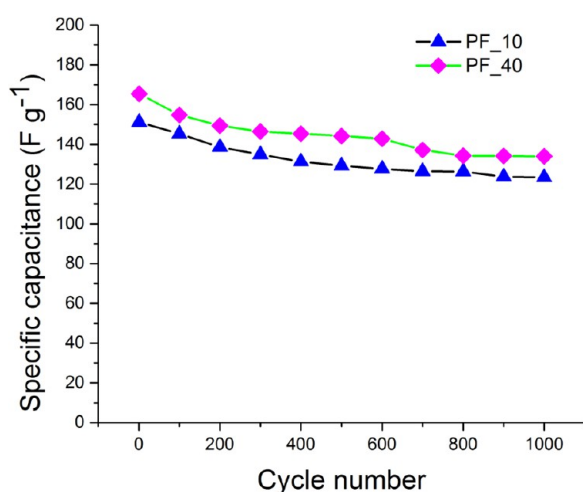


Figure 9. Cycling data for each 3D Om PF carbon electrode in a symmetrical cell. Cells were charged and discharged at 5.0 A g^{-1} (10 mA absolute current).

shapes characteristic of EDLC materials (SI Figure S9). The first and second charging curves featured a sluggish increase in cell voltage above 2.5 V. The ohmic drop for this sample is about 40 mV, which is of the same magnitude as values for the PF carbon materials. The calculated specific capacitance for the

activated carbon is about 155 F g^{-1} and comparable to the capacitance of 3D Om PF carbons.

Electrochemical Impedance Spectroscopy Studies.

Electrochemical impedance spectroscopy (EIS) was used to study the capacitive behavior and the ion-transport kinetics within mesopores and micropores of 3D Om PF carbons. The Nyquist plots in the frequency range from 100 Hz to 10 mHz of all 3D Om PF carbons feature a sharp increase of the imaginary part of the cell impedance at low frequencies, consistent with the good capacitive behavior of these samples (Figure 10). The zoomed-in regions in the high frequency range of the Nyquist plots show the first intercept of the real part to be near 3Ω for all 3D Om PF carbons. This value is typically the combined resistance of the electrolyte, the current collector, and the contact resistance between the electrode and the current collector. The suppressed semicircle in the high frequency range corresponds to a superposition of several phenomena, such as charge-transfer resistance and mass transport through mesopores and micropores in the 3D Om carbon materials. Figure 10B shows the frequency response of 3D Om PF carbons. The plots of capacitance vs frequency in the range from 10 mHz to 100 Hz for all 3D Om PF carbons have plateaus when the frequency approaches 0.1 Hz, indicating that the response time of 3D Om carbons is typically 10 s.

Cyclic Voltammetry Studies. Cyclic voltammetry (CV) was used to further characterize the electrochemical perform-

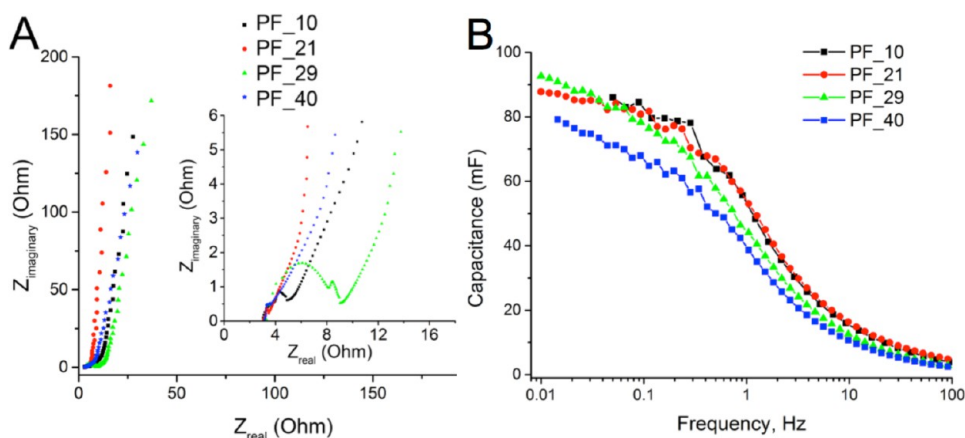


Figure 10. (A) Nyquist plots and (B) frequency responses of 3D Om PF carbons with different pore sizes.

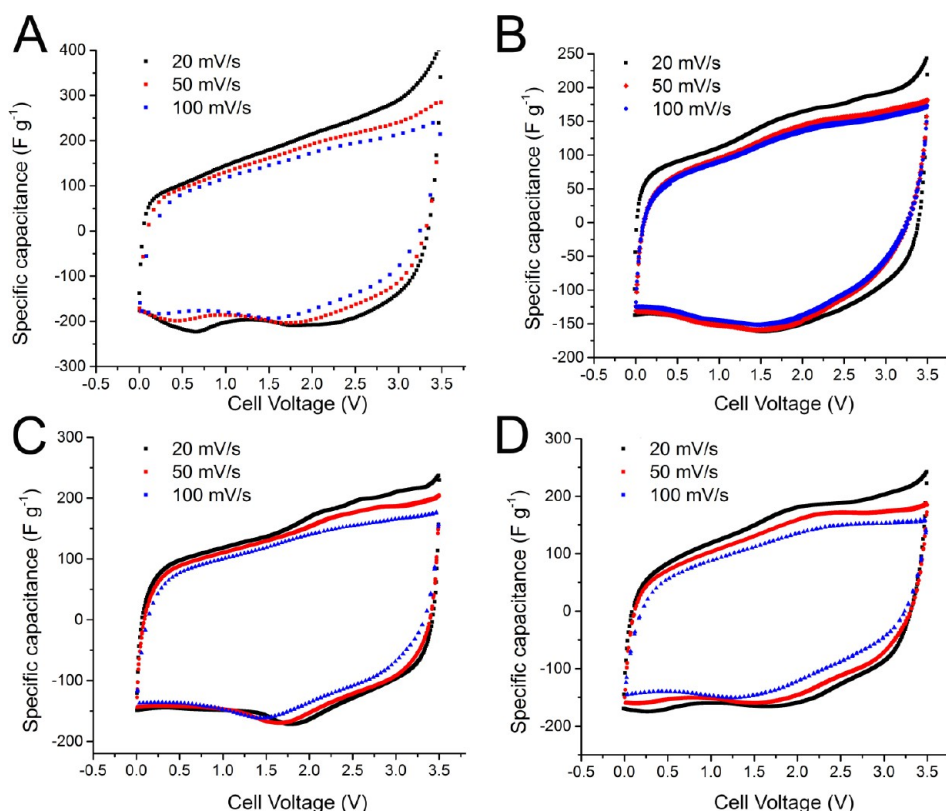


Figure 11. CV plots of 3D0m (A) PF_10, (B) PF_21, (C) PF_29, and (D) PF_40 carbons. The specific capacitance values are given with respect to the mass of carbon in each electrode of a symmetrical cell.

ance of 3D0m PF carbons. The cells were cycled between 0 and 3.5 V at different scan rates, ranging from 20 to 100 mV/s. The CVs of all 3D0m carbons exhibited rectangular shapes and good cyclability, characteristic for capacitors in which the charge/discharge rates are not significantly limited by ion diffusion (Figure 11). A sharp increase in the cathodic current at 3.2–3.5 V observed in most cases may be attributed to the oxidation of oxygen-containing functional groups on the surface of carbon. This feature slowly disappeared in the second and the following cycles. It may also be related to the stability limit of the electrolyte in the presence of trace amounts of water or oxygen. We observed that this problem was less prominent if a freshly opened EMI-TFSI bottle was used. When cycled at different scan rates, 3D0m carbons showed high rate capabilities with the capacitance decreasing only slightly upon a 5-fold increase in scan rate.

Rate Performance. The rate performance of 3D0m RF, 3D0m PF, and activated carbons was tested by performing charge and discharge cycles at different rates, ranging from 0.5 to 25 A g⁻¹. SI Figure S10 shows a typical galvanostatic test result for 3D0m PF_40 carbon being charged and discharged at different rates. The characteristic triangular shape was maintained even at a current as high as 50 mA (current density: 25 A g⁻¹). The ohmic drops increased at the higher currents; however, the resistance of the cell calculated from different discharge currents showed a constant value of 5 Ω. This clearly showed that the cell resistance was not raised by processes such as limited ion diffusion, even at the highest current tested. This behavior definitely benefitted from the interconnected pore system of 3D0m carbon.

The rate performance of 3D0m carbons of different pore sizes is plotted in Figure 12. All samples showed an extremely

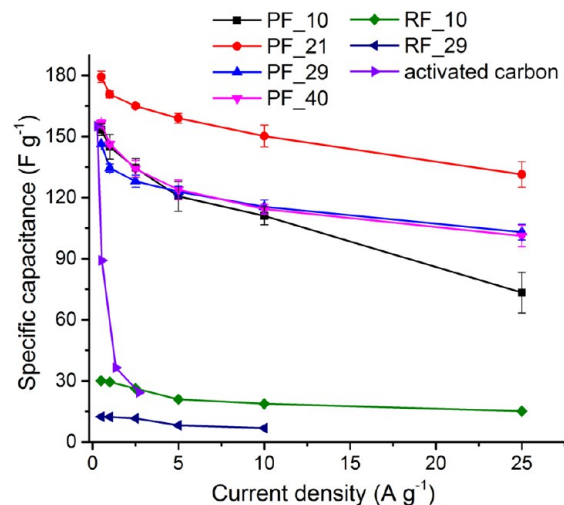


Figure 12. Plots comparing the rate performance of 3D0m PF, 3D0m RF, and activated carbons. The specific capacitance values are given with respect to the mass of carbon in each electrode of a symmetrical cell.

good rate performance with specific capacitance values ranging from 80 to 120 F g⁻¹ at the highest current density of 25 A g⁻¹. The performance of 3D0m PF carbons is remarkable, particularly for systems using an electrolyte with moderate conductivity such as EMI-TFSI. The rate performance of 3D0m PF_21 carbon is comparable to the most advanced systems based on hierarchical carbide-derived carbons and activated graphene, which showed 140–170 F g⁻¹ at a slow and 100–160 F g⁻¹ at a high rate.^{20,21,43} It should be noted that in

those reports, ionic liquids with higher ionic conductivity, including EMI-BF₄ and BMIM-BF₄, were used as electrolytes, and the best sample with about 160 F g⁻¹ was tested at a much smaller rate than the maximum rate used for 3DOm carbon here (5.7 A g⁻¹ compared to 25 A g⁻¹). All 3DOm PF carbons were able to maintain more than 50% of their capacitance when the current density was increased by a factor of 50 from 0.5 to 25 A g⁻¹.

It is interesting that the sample with medium-sized mesopores (PF_21) exhibits better rate performance than the samples with smaller (PF_10) or larger mesopores (PF_29 and PF_40). Since all of these samples have nearly the same ordered structure (cubic) and were prepared from the same precursor (PF sol), it is more likely that the mesopore sizes and the interactive roles of mesopores and micropores determine the rate performance of the materials. At small charge and discharge current densities, the performance is not affected by the diffusion of ions through the pores, and all 3DOm carbon samples show capacitance values of similar magnitude because of their similar surface areas. 3DOm PF_21 carbon with a slightly higher surface area has the highest specific capacitance of 178 F g⁻¹. The specific capacitance values of the other samples are about 150 F g⁻¹. At higher current densities, the diffusion of ions from the electrolyte becomes more important and affects the capacitor behavior. The interconnected pore system of the 3DOm structure benefits ion transport, resulting in the high capacitance at high rate. For the smallest mesopore sizes (i.e., for PF_10 carbon), the high rate performance is not as good as for the other PF carbons. The rate performance of the larger pore samples is decided by both mesopores and the micropores in the walls. The capacitance contribution from the micropores can be significant for these samples. If the mesopore walls are thin (PF_21), most of the micropores remain easily accessible to the electrolyte. However, as the wall thickness increases (PF_29 and PF_40), electrolyte transport through micropores becomes increasingly more restricted, leading to the small observed decrease in specific capacity for these materials at the highest current densities.

3DOm RF carbons have very small capacitance values (<30 F g⁻¹) and exhibit poor rate performance. In fact, 3DOm RF_29 carbon was not able to maintain a high current density (>10 A g⁻¹) due to the high resistance of the sample (Figure 12). Because of their poor performance, 3DOm RF carbons will not be discussed further.

The charge and discharge profile of activated carbon showed good capacitive behavior with triangular shape and a small ohmic drop at small current densities (SI Figure S11). However, at higher current densities, the triangular shape was strongly distorted by the large ohmic drop. This can be explained by the dominant micropores of activated carbon, which limit electrolyte accessibility, particularly at such high current densities. The capacitance of activated carbon dropped quickly upon increasing the current density (SI Figure S11B). The activated carbon-based cell was not able to maintain a current density higher than 2.7 A g⁻¹, and the voltage overshoot the set values (<0 V or >3.5 V). This provides strong evidence for the advantage of having an ordered porous structure as in 3DOm carbon, minimizing ion diffusion limitations at high current densities.

Effect of Sample Mass. It is known that the performance of an EDLC cell depends strongly on the thickness of the electrodes, thicker electrodes normally exhibiting poorer power capability. Therefore, depending on the targeted application of

the EDLC cells, they are assembled with different amounts of active materials. In general, thin electrodes are used for high power cells and thick electrodes for high energy cells.³¹ Cells with both high power density and high energy density have been long-standing targets of research. However, this is difficult to achieve with activated carbon, the most common electrode material for EDLCs, due to its limited rate performance when thick electrodes are used. 3DOm carbons may provide a solution to this problem because of their comparatively high capacitance and their outstanding rate performance.

To study the effect of electrode thickness on the power capability of 3DOm PF carbons, electrodes with two different loadings of active material, that is, 4.4 ("thin") or 26 mg/cm² ("thick"), were prepared. The rate performance of 3DOm PF carbon with small mesopores (PF_10) was strongly affected by the thickness of the electrode, with a decrease in specific capacitance of more than 50% when the loading was increased from 4.4 to 26 mg/cm² (Figure 13). In contrast, the rate

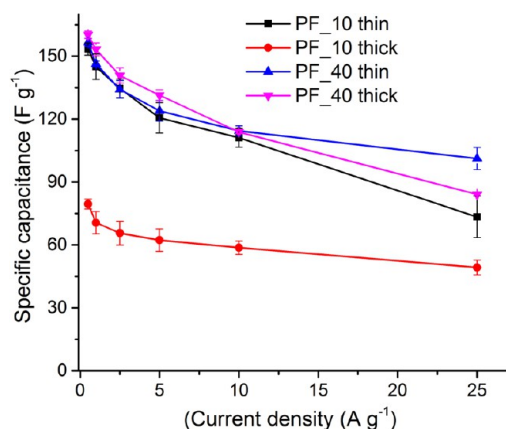


Figure 13. Plots comparing the rate performance of 3DOm PF carbon electrodes prepared with different mass loadings of active material. The specific capacitance values are given with respect to the mass of carbon in each electrode of a symmetrical cell.

performance of 3DOm PF carbon with large mesopores (PF_40) decreased only slightly (<20%) with the same increase in electrode loading. This suggests that 3DOm carbons with interconnected, large mesopores are more suitable to build EDLC cells with combined high power and high energy densities.

CONCLUSIONS

3DOm carbons, prepared by colloidal crystal templating, with ordered, interconnected mesopores in the 10–40 nm range and accessible micropores in the walls, were demonstrated to provide special advantages for electrodes in EDLCs used with ionic liquid electrolytes. The microporous walls provide the high surface areas needed for high capacitance, while the interconnected mesopore system is necessary to enable efficient ion transport for high power capabilities. Fine control over pore sizes was possible through colloidal crystal templating by adjusting the size of silica sphere templates via the hydrolysis temperature and the amount of TEOS during seed growth. All 3DOm carbon materials prepared from PF or RF precursors showed ordered structures and high surface areas of 1000–1200 m² g⁻¹ for PF carbons and up to 1600 m² g⁻¹ for RF carbons. However, the 3DOm PF and RF carbons performed very differently in electrochemical tests, including galvanostatic

cycling, electrochemical impedance spectroscopy, and cyclic voltammetry. In all tests, 3DOm PF carbons showed much better performance than RF carbons, which can be ascribed to the different oxygen content in the precursors, influencing the conductivity. The best 3DOm PF carbons showed high specific capacitance (178 F g^{-1} at 0.5 A g^{-1} current density) and high rate performance (123 F g^{-1} at 25 A g^{-1} current density). The performance of 3DOm carbons is comparable to some of the most advanced systems reported in the literature.

■ ASSOCIATED CONTENT

■ Supporting Information

Photograph of EDLC cell components, comparison of theoretical and measured sphere sizes, additional TEM and SEM images, Raman data, galvanostatic data for 3DOm RF carbons and activated carbon. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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